Emergency Department Wait Time Prediction: Systematic Literature Review

Milestone 2 – Data Collection, Wrangling & Visualization

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Data Science & Machine Learning

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From Literature → Data

M1 — Gaps We Found

- Most studies use binary classification only.
- No triage-level (ESI 1-5) segmentation reported.
- Limited SHAP interpretability (4 of 12 studies).
- 9 of 12 use private, single-center datasets.
- Only 2 of 12 address demographic fairness.

→

M2 — Our Response

- Continuous regression — predict LOS in minutes.
- Kaggle dataset with all 5 ESI levels.
- SHAP-ready ML pipeline planned for M3.
- Kaggle public dataset (CCO) — fully reproducible.
- Fairness analysis across race, gender, age.

Dataset: Hospital Emergency Room Visits

9,216
ED Visits

26
Features

189.9
Mean LOS (min)

CC0
Public License

Feature	Type	Description
age	Numeric	Patient age (years)
heartrate	Numeric	Heart rate (bpm)
sbp / dbp	Numeric	Systolic / diastolic BP
temperature	Numeric	Body temp (°F)
o2sat	Numeric	Oxygen saturation (%)
pain_score	Numeric	Pain 0–10
acuity	Ordinal	ESI level 1–5

Engineered Feature	Description
shock_index	HR / SBP ratio
pulse_pressure	SBP – DBP
arrival_shift	Day / Evening / Night
is_weekend	Binary weekend flag
age_group	Pediatric / Young Adult / etc.
ed_los_minutes ★	**TARGET — continuous LOS in minutes**

Source: Kaggle — drnimishadavis · Accessed March 2026 · CC0 Public Domain

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Data Wrangling Pipeline

Part B — 5 sequential steps applied to all 9,216 records

1 Missing Values

7 vital-sign cols had 8–12% missing
Median imputation (vitals) + mode (categorical)

2 Type Conversion

Label encoding for gender, race, arrival_transport, arrival_shift, age_group

3 Outlier Handling

IQR + Z-score detection. Clinical bounds verified — 0 values required capping

4 Feature Engineering

5 new features:
shock_index
pulse_pressure
arrival_shift
is_weekend (proxy)
age_group

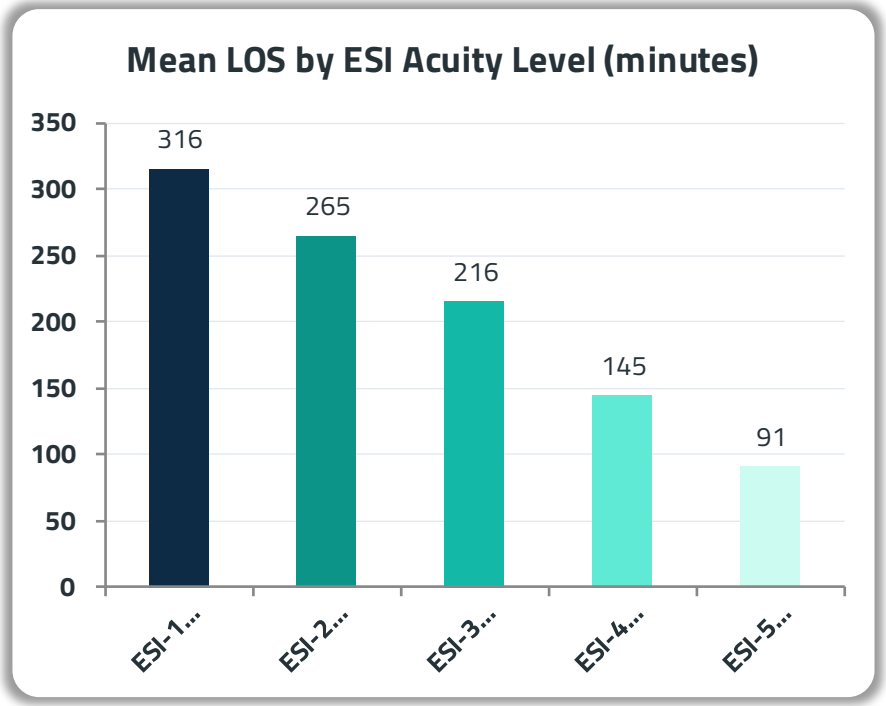
5 Normalization

StandardScaler on 19 numeric features
Target ed_los_minutes kept raw (not scaled)

9,216 clean records · 17 → 28 cols after wrangling (5 engineered + 6 encoded) · 5,156 missing → 0 · Target: ed_los_minutes (8.6 – 2,644.9 min)

EDA: Target Variable & Acuity

Part C — Acuity (ESI level) is the dominant predictor of ED length of stay



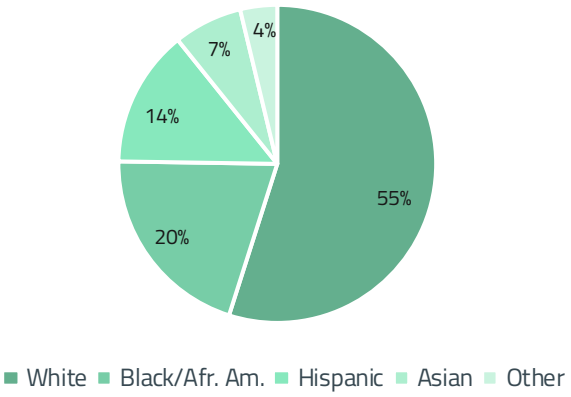
153.7	Median LOS (min)
189.8	Mean LOS (min)
r =−0.38	Acuity↔LOS Correlation
4,184	ESI-3 patients (45%)

Key insight: Acuity (r = −0.38) is the only meaningful predictor. ESI-1 patients stay 3.5× longer than ESI-5. All vital signs show r ≈ 0.00–0.02.

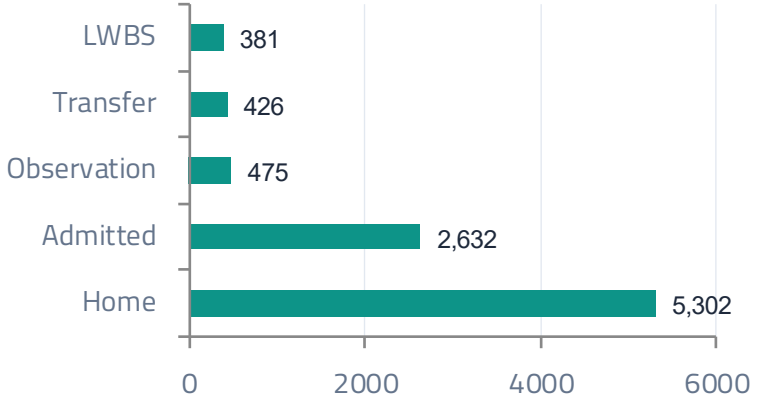
EDA: Demographics & Disposition

Part C — Race distribution, patient outcomes, and fairness preview

Patient Race Distribution



Disposition Outcomes

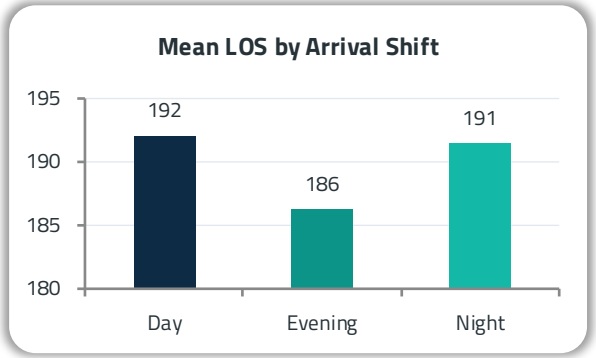


Fairness Preview — Mean LOS (min) by demographic subgroup:

188.8	190.9	193.5	179.2	182.0
Female	Male	Young Adult	Elderly	Pediatric

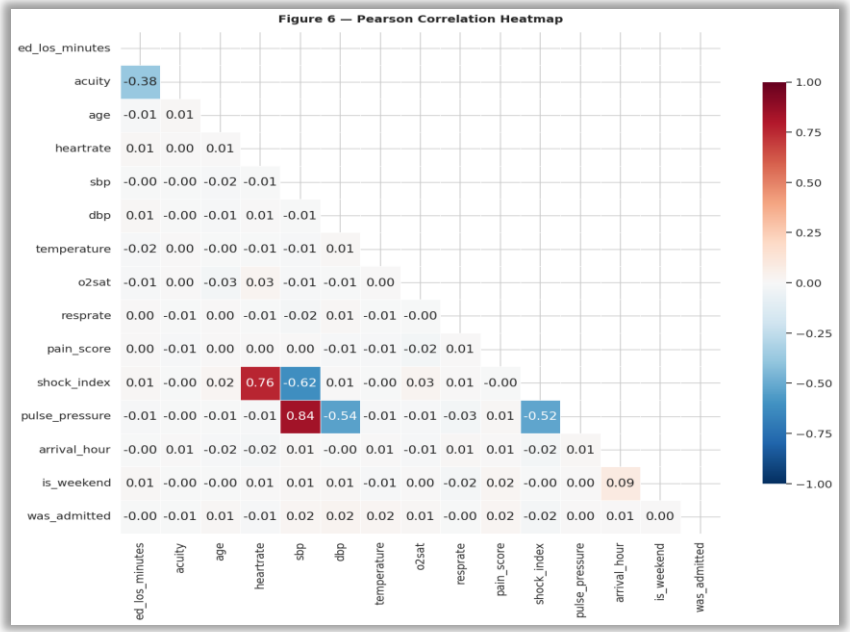
EDA: Temporal Patterns & Engineered Features

Part C — Shift analysis, engineered features, and Pearson correlation heatmap (Figure 6)



Insight: Shift has minimal effect on LOS (< 6 min range) — operational crowding is captured more by acuity than time of day.

Engineered Features		
shock_index	HR/SBP	Hemodynamic instability proxy
pulse_pressure	SBP-DBP	Cardiovascular risk marker
arrival_shift	Hour → Day/Eve/Night	Operational congestion signal
is_weekend	Arrival Weekday flag	Staffing level proxy
age_group	Age → 4 categories	Clinical complexity proxy



Key finding: Only acuity shows meaningful correlation. Vitals show r ≈ 0.00–0.02, confirming that ensemble models.
Capturing non-linear feature interactions are essential for M3. SBP↔ DBP are strongly correlated (multicollinearity → motivates pulse_pressure).

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ML Pipeline – Intermediate Setup

Part C — M3 roadmap: 80/20 train/test split · Stratified · Target: continuous_ed_los_minutes

Cleaned Dataset
9,216 rows
26 features

→

Train / Test Split
80% / 20%
Stratified

→

Feature Matrix X
19 numeric
+ encoded cols

→

Target Vector y
ed_los_minutes
continuous (min)

→

M3 — Model Training
XGBoost
RF · LightGBM

XGBoost

- Primary model.
- Handles non-linearity well.
- Built-in feature importance.
- Best literature baseline.

Random Forest

- Robust ensemble baseline.
- Resistant to outliers.
- Comparable to Wang et al. (2025).
- Easily interpretable.

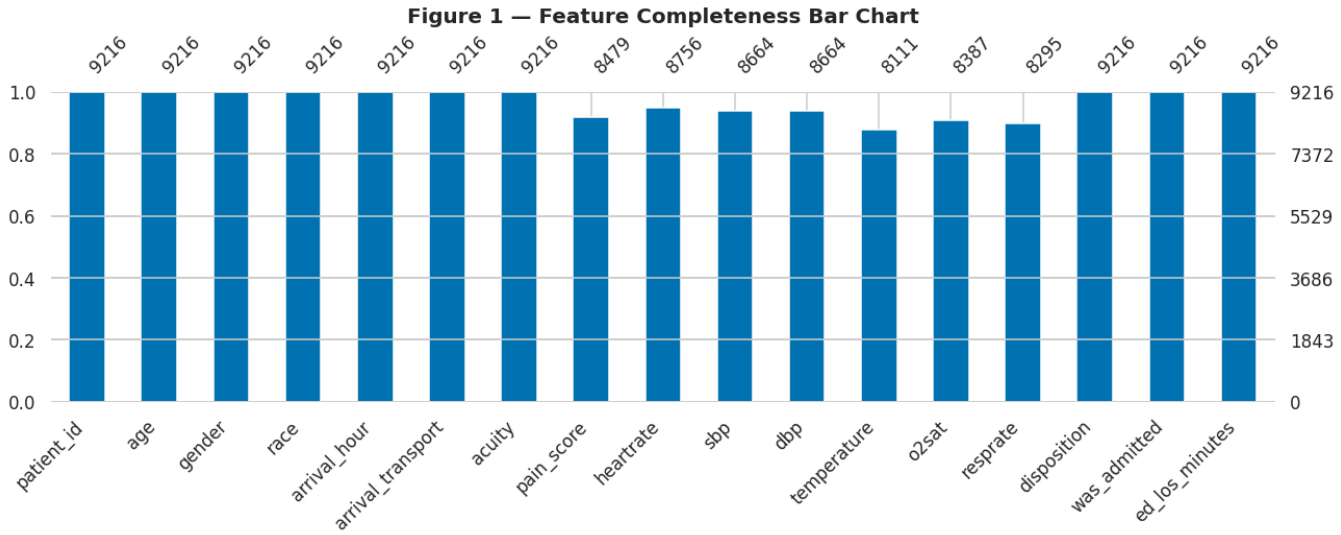
LightGBM

- Fastest training speed.
- Best performer in Wu et al. (2025).
- Lowest MAE on continuous LOS.
- Memory efficient.

Evaluation Metrics (M3): MAE · RMSE · R^2 · SHAP Feature Importance · ESI-Level Subgroup Performance · Fairness by Race/Gender/Age

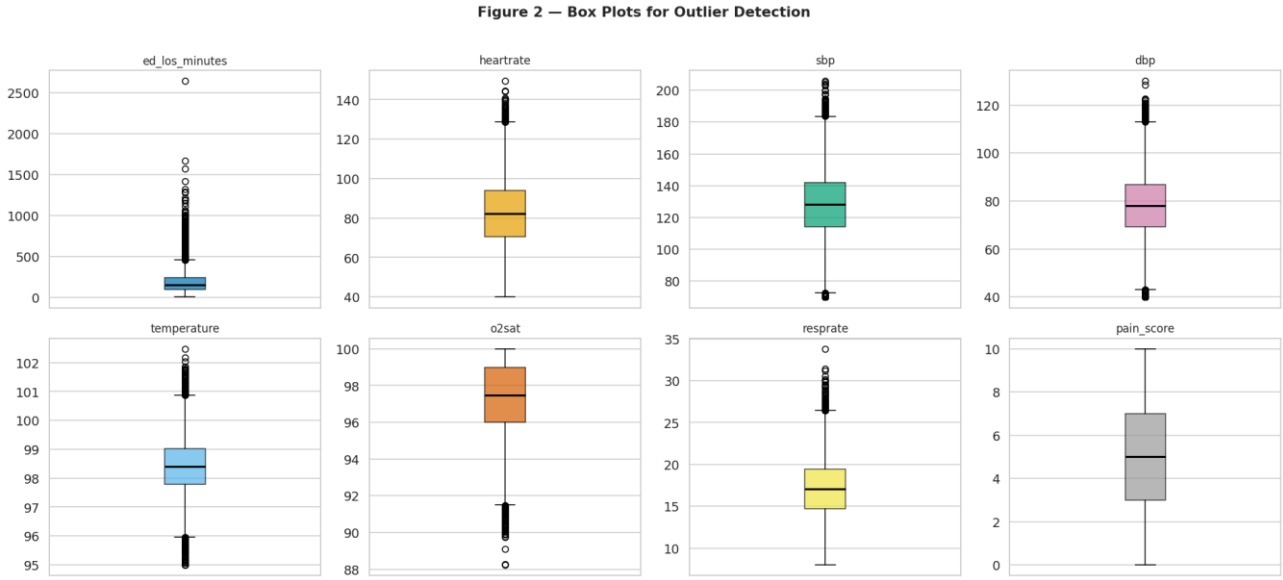
Notebook Highlights – Code & Results

Part B — Figure 1- Feature completeness before imputation



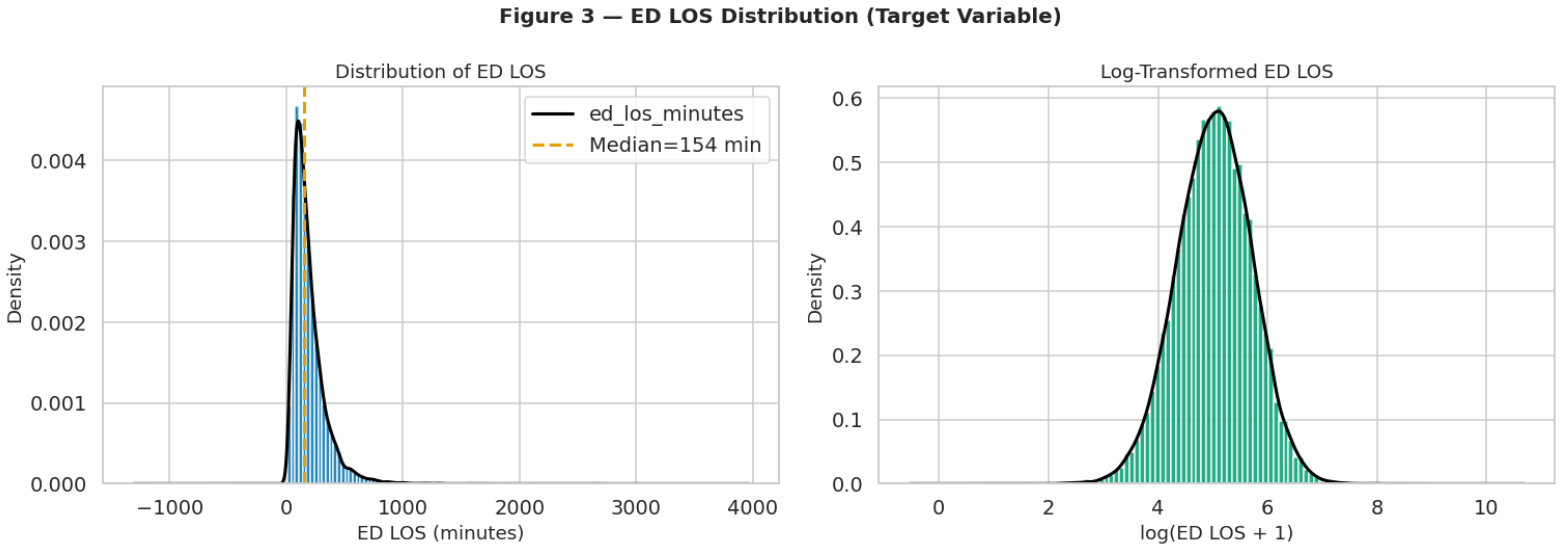
Interpretation: 7 vital-sign columns had 8–12% missing values prior to wrangling (temperature 12%, resprate 10%, o2sat 9%, pain_score 8%, sbp/dbp 6%, heartrate 5%). This is expected in ED triage settings where not all vitals are measured for every patient. No column exceeded 15% — all were retained and imputed using median for numerics and mode for categoricals, resulting in 0 missing values in the final cleaned dataset.

Part B — Figure 2 · IQR + Z-score boxplots for all numeric features



Interpretation: ed_los_minutes shows the most outliers by IQR (4.5% of records, values up to 2,645 min) — expected for complex boarding cases in real ED settings. Vital signs (heartrate, sbp, dbp, temperature, o2sat, resprate) show medically plausible distributions with limited extremes. Clinical bounds verification confirmed 0 values required capping: all vitals were already within defined medical ranges (e.g., heartrate 30–250, sbp 60–280). IQR and Z-score methods were applied for detection but winsorization had no effect, confirming high data quality.

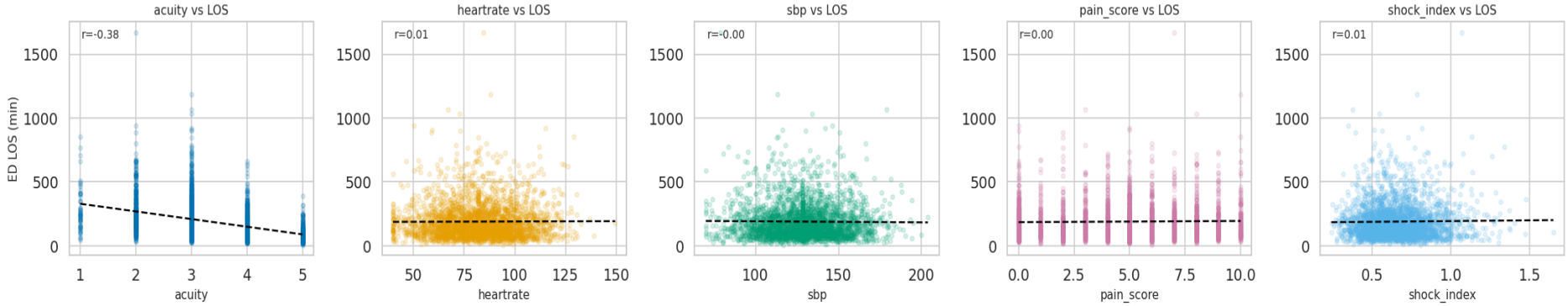
Part C — Figure 9 · Key features vs. ed_los_minutes with Pearson r



Interpretation: ED LOS is strongly right-skewed (median = 154 min, mean = 189.8 min, max = 2,644.9 min), with most visits under 600 minutes. The long tail represents complex boarding cases — a clinically realistic pattern consistent with Wu et al. (2025). The log-transformed distribution approaches normality, suggesting log-transformation of the target variable may improve regression model performance in Milestone 3. Gradient boosting models (XGBoost, LightGBM) handle skewed targets well without transformation, but this will be evaluated empirically.

Part C — Figure 9 · Key features vs. ed_los_minutes with Pearson r

Figure 9 — Scatter Plots: Key Features vs. ED LOS



Interpretation: Acuity is the only feature with a meaningful negative linear relationship with LOS (r = −0.38): higher ESI number = lower severity = shorter stay. Heartrate (r = 0.01), SBP (r = −0.00), pain_score (r = 0.00), and shock_index (r = 0.01) all show near-zero Pearson correlations, confirming that vital signs have no significant linear relationship with LOS individually. The wide scatter across all plots confirms ED LOS is a multivariate, non-linear problem — exactly why ensemble tree models that capture feature interactions are the right modeling approach for Milestone 3.

Milestone 3 – Next Steps

M2 — Completed

- 9,216 ED visits collected & cleaned
- 5,156 missing values resolved → 0 remaining.
- 5 features engineered from clinical signals.
- 12 EDA visualizations with interpretations.
- ML-ready CSV (26 columns) exported.

→

M3 — Roadmap

1. Train XGBoost, Random Forest, LightGBM.
2. Evaluate: MAE, RMSE, R^2.
3. SHAP feature importance (patient + population level).
4. ESI triage-level subgroup performance analysis.
5. Demographic fairness: race, gender, age group.

Research Question: Which combination of patient and operational variables best predicts ED LOS, and which model balances predictive performance with clinical interpretability?

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